CSL100/MAL100

Tutorial 6

Function and array

Question 1:

Write a program in C to calculate the sum of numbers from 1 to n using recursion.

Example :
Input the last number of the range starting from 1 : 5

Expected Output:
The sum of numbers from 1 to 5 :
15

```c
#include <stdio.h>

// Recursive function to find the sum of n natural numbers
int sumNaturalNumbers(int n) {
    if (n == 0)
        return 0;
    return n + sumNaturalNumbers(n - 1);
}
```

```c
int main() {
    int n;


    printf("Enter the value of n: ");
    scanf("%d", &n);

    printf("The sum of the first %d natural numbers
is: %d\n", n, sumNaturalNumbers(n));

    return 0;
}
```

Question 2:

Write a C program using recursion to print the Fibonacci series up to n terms.

Example:
I/P : enter number = 5
O/P: Fibonacci series 5th term= 0 1 1 2 3

```c
#include <stdio.h>

// Recursive function to calculate nth Fibonacci number
int fibonacci(int n) {
    if (n == 0)
        return 0;
    if (n == 1)
        return 1;
    return fibonacci(n - 1) + fibonacci(n - 2);
}

// Function to print Fibonacci series up to n terms
void printFibonacciSeries(int n) {
    for (int i = 0; i < n; i++) {
        printf("%d ", fibonacci(i));
    }
}
```

```c
int main() {
    int n;

    // Input number of terms
    printf("Enter number of terms: ");
    scanf("%d", &n);

    // Output Fibonacci series up to n terms
    printf("Fibonacci series up to %dth term: ", n);
    printFibonacciSeries(n);

    return 0;
}
```

Question 3.

Write a C program using recursion to find the sum of even and odd numbers up to n terms.

Example:
Input: Enter the nth term : 10
Output: The sum of odd numbers is 25 .
The sum of even numbers is 30

```c
#include <stdio.h>

int sumEven(int n) {
    if (n <= 0)
        return 0;
    if (n % 2 == 0)
        return n + sumEven(n - 2);
    return sumEven(n - 1);
}



int sumOdd(int n) {
    if (n <= 0)
        return 0;
    if (n % 2 != 0)
        return n + sumOdd(n - 2);
    return sumOdd(n - 1);
}
```

```c
int main() {
    int n;

    // Input nth term
    printf("Enter the nth term: ");
    scanf("%d", &n);


    int oddSum = sumOdd(n);
    int evenSum = sumEven(n);

    printf("The sum of odd numbers is
%d.\n", oddSum);
    printf("The sum of even numbers is
%d.\n", evenSum);

    return 0;
}
```

Question 5

Write a program in C to store elements in an array and print them.

Example :
Input 10 elements in the array :
element - 0 : 1
element - 1 : 1
element - 2 : 2
.....
Expected Output :
Elements in array are: 1 1 2 3 4 5 6 7 8 9

```c
#include <stdio.h>

int main() {
    int arr[10];  // Declare an array to hold 10 elements
    int i;



       printf("Input 10 elements in the array:\n");
    for(i = 0; i < 10; i++) {
        printf("element - %d : ", i);
        scanf("%d", &arr[i]);
    }



    printf("Elements in array are: ");
    for(i = 0; i < 10; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

Question 5

Write a C program that creates an array named student_id to store at least 5 student IDs. Implement a function named search that takes the array and a student ID as input and checks whether the given ID is present in the array. The function should return 1 if the ID is found and 0 if the ID is not found.
Example:
Enter 5 student IDs:
ID 1: 121
ID 2: 212
ID 3: 1223
ID 4: 123
ID 5: 43
Expected output: Enter the student ID to search: 56
Student ID not found!
Expected output:
Enter the student ID to search: 43
Student ID found

```c
#include <stdio.h>

// Function to search for a student ID in the array
int search(int student_id[], int size, int id) {
    for (int i = 0; i < size; i++) {
        if (student_id[i] == id)
            return 1;  // ID found
    }
    return 0;  // ID not found
}
```

```c
int main() {
    int student_id[5], idToSearch;

    printf("Enter 5 student IDs:\n");
    for (int i = 0; i < 5; i++) {
        printf("ID %d: ", i + 1);
        scanf("%d", &student_id[i]);
    }

    // Input the student ID to search
    printf("Enter the student ID to search: ");
    scanf("%d", &idToSearch);


    if (search(student_id, 5, idToSearch)) {
        printf("Student ID found!\n");
    } else {
        printf("Student ID not found!\n");
    }


    return 0;
}
```

QUIZ

1. Write a C program to read at least 5 integers from the user and store them in an array. Implement a recursive function to find the maximum value in the array, which takes the array and its size as arguments and returns the largest element.

   Example: Enter 5 numbers: 15 marks
   Number 1: 10
   Number 2: 25
   Number 3: 5
   Number 4: 30
   Number 5: 15
   Expected Output: The maximum value in the array is: 30